

Grade 6 Accelerated
Unit 13
Lesson 2 Homework

Name _____
Date _____
Period _____

1. Evaluate the expression $8b + 10 - (9 - 6)^2$ for $b = -3$.

$$\begin{aligned} & 8(-3) + 10 - (9 - 6)^2 \\ & -24 + 10 - 3^2 \\ & -24 + 10 - 9 \\ & -14 - 9 \\ & = -23 \end{aligned}$$

2. Evaluate $-7m - 5q^2 + \frac{s}{2}$ when $m = 8$, $q = -2$, and $s = -48$.

$$\begin{aligned} & -7(8) - 5(-2)^2 + \frac{-48}{2} \\ & -56 - 5(4) - 24 \\ & -56 - 20 - 24 \\ & -76 - 24 \\ & = -100 \end{aligned}$$

3. Evaluate $2s - 6q - f^2$ for $s = 4$, $q = -6$, and $f = -5$.

$$\begin{aligned} & 2(4) - 6(-6) - (-5)^2 \\ & 8 + 36 - 25 \\ & 44 - 25 \\ & = 19 \end{aligned}$$

4. Evaluate $-m^3 - 3d + (n - y)$ if $m = 3$, $d = 7$, $n = 9$, and $y = 1$.

$$\begin{aligned} & -3^3 - 3(7) + (9 - 1) \\ & -27 - 21 + 8 \\ & -48 + 8 \\ & = -40 \end{aligned}$$

5. Evaluate $4g^4 - \frac{z}{4}$ when $g = 2$ and $z = 16$.

$$\begin{aligned} & 4(2)^4 - \frac{16}{4} \\ & 4(16) - 4 \\ & 64 - 4 \\ & = 60 \end{aligned}$$

6. What is the value of $5(8x + 4) - 3m^2$ if $x = -3$ and $m = 5$?

$$\begin{aligned} &5(8(-3) + 4) - 3(5)^2 \\ &5(-24 + 4) - 3(25) \\ &5(-20) - 3(25) \\ &-100 - 75 \\ &= -175 \end{aligned}$$

7. Find the value of $-9(h - 6) + 2a - (9 - a)^5$ if $h = 8$, and $a = 7$.

$$\begin{aligned} &-9(8 - 6) + 2(7) - (9 - 7)^5 \\ &-9(2) + 2(7) - 2^5 \\ &-18 + 14 - 32 \\ &-4 - 32 \\ &= -36 \end{aligned}$$

8. Evaluate the expression $p^3 - 5k(y - k)$ for $p = -4$, $k = 6$ and $y = 10$.

$$\begin{aligned} &(-4)^3 - 5 \cdot 6(10 - 6) \\ &-64 - 30(4) \\ &-64 - 120 \\ &= -184 \end{aligned}$$

9. Evaluate the expression $8 \div j - 12 \div (-10 + 4)$ if $j = 2$.

$$\begin{aligned} &8 \div 2 - 12 \div (-10 + 4) \\ &8 \div 2 - 12 \div (-6) \\ &4 - (-2) \\ &= 6 \end{aligned}$$

10. What is the value of the expression $4m - 10 + (a - 11) \cdot 4^3$ if $m = 9$ and $a = 5$?

$$\begin{aligned} &4(9) - 10 + (5 - 11) \cdot 4^3 \\ &4(9) - 10 + (-6) \cdot 64 \\ &36 - 10 - 384 \\ &26 - 384 \\ &= -358 \end{aligned}$$